1 equations

ALL IN THIS 1-2:

$\frac{A_2}{B_2} = 8\pi\bar{h}\frac{\mu^3}{\lambda^3}$;

$g_1 B_{12} = g_2 B_{21}$

$-dn_2(\nu) = A_{21} n_2 f(\nu) d\nu$;

$-dn_2 = B_{21} n_2 f(\nu) \rho_\nu d\nu$

$f(\nu) = \frac{\frac{\Delta\nu}{2\pi}}{(\nu-\nu_0)^2+(\frac{\Delta\nu}{2})^2}$;

$\Delta\nu = \frac{1}{2\pi\tau}$;

GAP= $\sqrt{1+\frac{I}{I_s}}\Delta\nu$;

Stable $\boldsymbol{Condition}$:$0 < (1-\frac{L}{R_1})(1-\frac{L}{R_2}) < 1$;

$\Delta n = \frac{[(\nu-\nu_0)^2+(\frac{\Delta\nu}{2})^2]\Delta n_0}{(\nu-\nu_0)^2+(1+\frac{I}{I_s})(\frac{\Delta\nu}{2})^2} = \frac{\Delta n_0}{1+\frac{I}{I_s}\frac{f(\nu)}{f(\nu_0)}}$;

$\Delta n = n_2 - \frac{g_2}{g_1} n_1$;

$G = \Delta n B_{21} f(\nu) h\nu \frac{mu}{c}$;

$\Delta n = R_2\tau_2 - (R_1+R_2)\tau_1$;

$G = \frac{[(\nu-\nu_0)^2+(\frac{\Delta\nu}{2})^2]G^0}{(\nu-\nu_0)^2+(1+\frac{I}{I_s})(\frac{\Delta\nu}{2})^2} = \frac{G^0}{1+\frac{I}{I_s}\frac{f(\nu)}{f(\nu_0)}}$;

$G^0(\nu) = \Delta n_0 B_{21}\frac{\mu}{c} f(\nu) h\nu_0$

$\Delta n(\nu) = \frac{[(\nu-\nu_0)^2+(\frac{\Delta\nu}{2})^2]\Delta n_0 f_D(\nu)}{(\nu-\nu_0)^2+(1+\frac{I}{I_s})(\frac{\Delta\nu}{2})^2} = \frac{\Delta n_0 f_D(\nu)}{1+\frac{I}{I_s}\frac{f(\nu)}{f(\nu_0)}}$;

wide of hole$\nu \pm \sqrt{1+\frac{I}{I_s}}\frac{\Delta\nu}{2}$

$G_D(\nu_1) = \frac{G_d^0(\nu_1)}{\sqrt{1+\frac{I}{I_s}}}$

$a_{all} = a_{in} - \frac{1}{2L} ln r_1 r_2 = -\frac{1}{2L} ln(1-(a_1+t_1))$

$\boldsymbol{Condition}$:$2\delta\Phi = 2q\pi, q = 1, 2, 3...$

$\Delta\nu_q = \frac{c}{2\mu L}$

$2L = q\lambda$

$\boldsymbol{s}quare\Delta\Phi_{mn} = (m+n+1)\frac{\pi}{2}$

$\nu_{nmq} = \frac{c}{2\mu L}(q+\frac{1}{2}(m+n+1))$

$\frac{1}{2}\Delta\nu_q = \Delta\nu_m = \Delta\nu_n$

$\boldsymbol{r}ound\Delta\Phi_{mn} = (m+2n+1)\frac{\pi}{2}$

$\nu_{nmq} = \frac{c}{2\mu L}(q+\frac{1}{2}(m+2n+1))$

$\frac{1}{2}\Delta\nu_q = \Delta\nu_m = \frac{1}{2}\Delta\nu_n$

$\boldsymbol{G}auss$:

$\omega(z) = \omega_0\sqrt{1+(\frac{f}{z})^2}$

$\omega_0 = \sqrt{\frac{\lambda f}{\pi}}$

$R(z) = z[1+(\frac{f}{z})^2]$

$L = |z_1| + |z_2|$

$R_1 = R_2 = 2f$(S.C chamber) not suitable for normal shpere.

$f = \frac{\pi\omega_0^2}{\lambda}$

$2\theta = \frac{2\lambda}{\pi\omega_0}$

$q = z + if$
$2\theta_{mn} = \sqrt{2m+1}\sqrt{2n+1}2\theta_0$
$V_{00}{}^0 = \frac{L^2\lambda}{2}$
$V_{mn}{}^0 = \sqrt{2m+1}\sqrt{2n+1}V_{00}{}^0$
HOMOS
$G = \frac{G^0}{1+2\frac{I}{I_s}} = G_{Thres}$
$a_{all} = \frac{a_1+t_1}{2L}$
$I = \frac{I_s}{2}t_1(\frac{2LG^0}{a_1+t_1} - 1)$
$P = IS$
non-HOMOS
non-center franquency :$G(\nu) = \frac{G_D^0(\nu)}{sqrt1+\frac{I}{I_s}} = G_{Thres}$
center franquency :$G(\nu) = \frac{G_D^0(\nu)}{sqrt1+2\frac{I}{I_s}} = G_{Thres}$